

Customer Segmentation & Retention Analytics

RFM Analysis, Customer Clustering, and Next-Purchase Prediction

Executive Summary: This project analyzes transaction-level retail data from the Online Retail II dataset to identify customer behavior patterns, build RFM-based segments, validate those segments using clustering, and estimate when active customers are likely to buy again. The output is a practical customer strategy framework that helps prioritize loyalty, reactivation, and win-back actions.

Problem Statement

Many businesses treat all customers the same, even though customers differ widely in how recently they bought, how often they buy, and how much they spend. This one-size-fits-all approach makes it harder to identify loyal customers, predict churn risk, and design targeted retention campaigns.

Objective

- Segment customers using Recency, Frequency, and Monetary (RFM) analysis.
- Identify high-value, loyal, potential loyalist, at-risk, and inactive customer groups.
- Validate behavioral patterns using K-Means clustering.
- Analyze product preferences by customer segment.
- Predict the number of days until the next purchase for active customers.
- Translate the findings into practical retention and growth recommendations.

Dataset and Data Preparation

The analysis is based on the Online Retail II transaction dataset. Two CSV files were combined into a single table of invoice-level transactions. The data was cleaned by removing missing Customer IDs, excluding cancelled orders, filtering out non-positive quantities, and converting invoice dates into a usable datetime format. A TotalPrice field was created from Quantity multiplied by Price.

Methodology

1. RFM Scoring

Each customer was assigned three core metrics: Recency (days since last purchase), Frequency (unique invoices), and Monetary Value (total spend). Quartile-based scoring was then used to assign R, F, and M scores, which were combined into an overall RFM score.

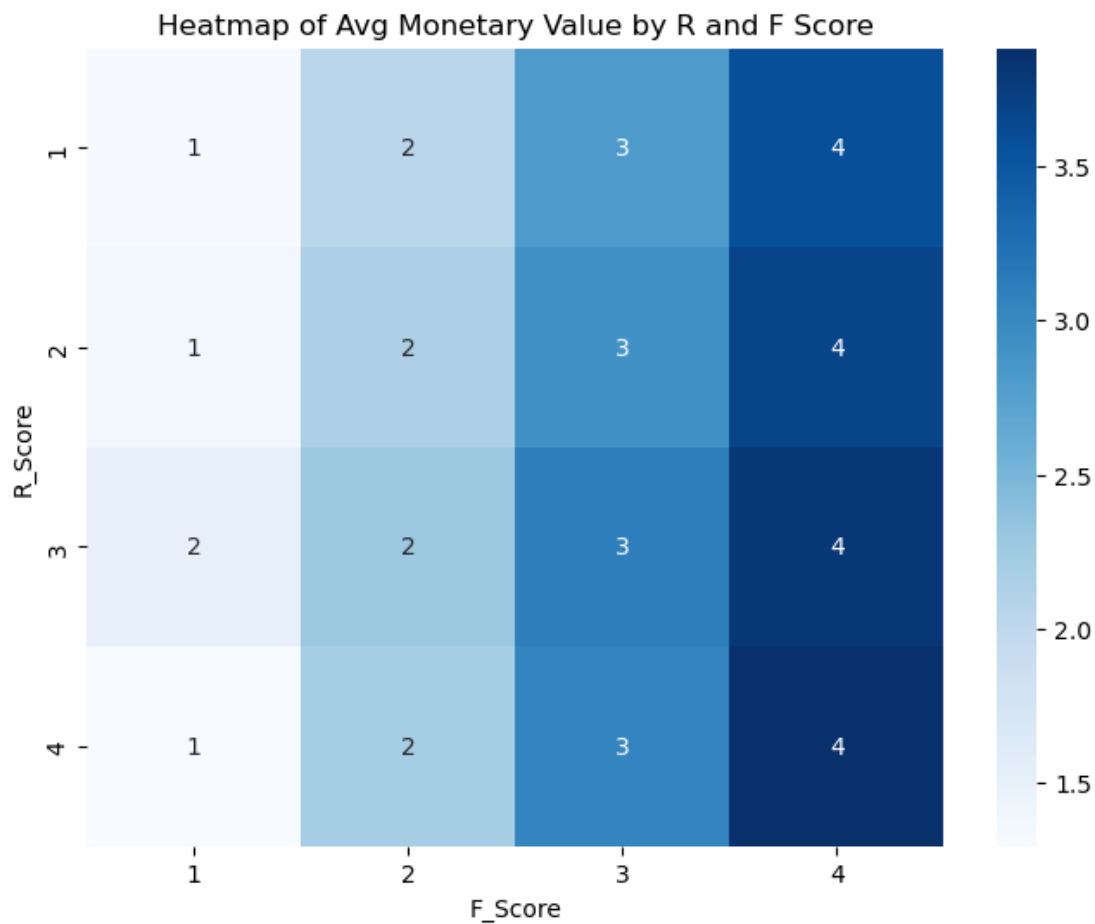


Figure 1. Heatmap of average monetary value by RFM score.

2. Customer Segmentation

Customers were grouped into business-friendly segments based on their RFM score: Champions, Loyal Customers, Potential Loyalists, At-Risk Customers, and Can't Lose Them. This makes the analysis easier to use for retention and lifecycle marketing decisions.

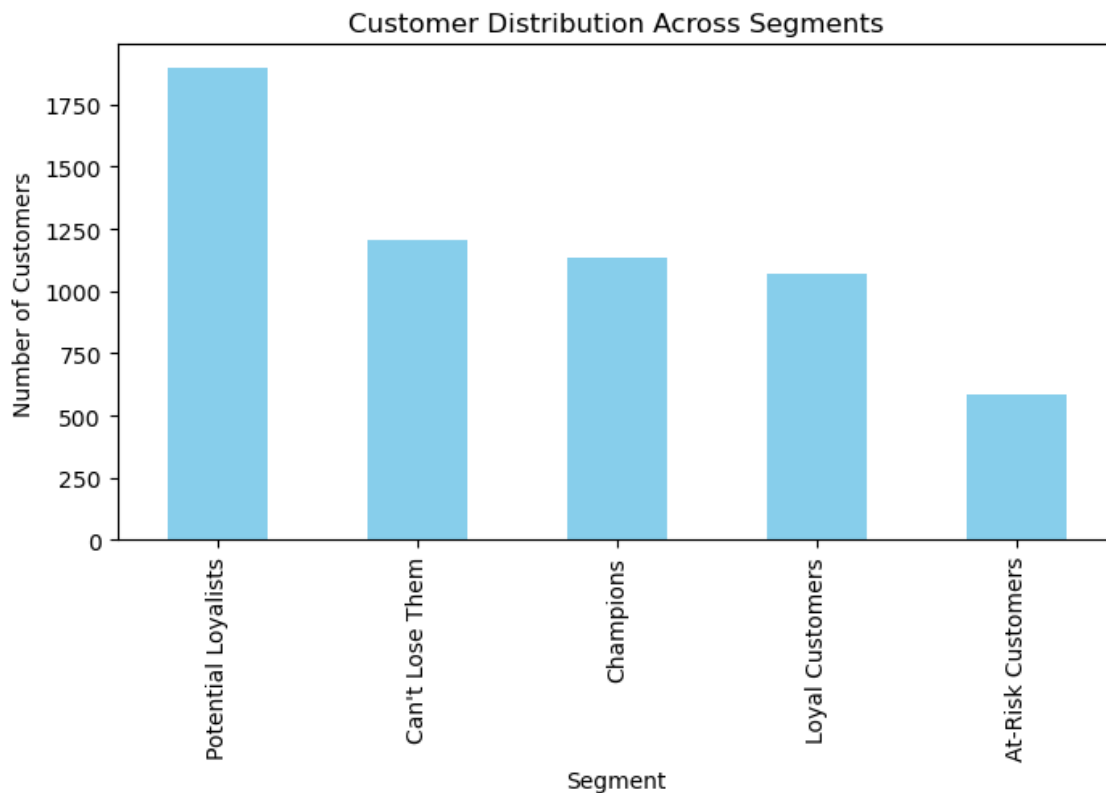


Figure 2. Distribution of customers across RFM segments.

3. Clustering Validation

To validate the rule-based segmentation, the RFM variables were log-transformed, standardized, and clustered using K-Means. The elbow method supported a four-cluster solution. The clusters closely aligned with the business segments and helped confirm the underlying customer structure.

4. Product Preference Analysis

Customer purchase descriptions were analyzed to identify what each segment tends to buy. This added a second layer of insight beyond spend and frequency by showing how product preferences differ across customer groups.

5. Next-Purchase Prediction

A Random Forest Regressor was used to estimate the number of days until the next purchase. The model used Recency, Frequency, Monetary Value, and average inter-purchase time as features. This produces a practical signal for timing re-engagement campaigns.

Key Findings

The average behavior of each business segment is summarized below.

Segment	Recency (days)	Frequency (orders)	Monetary Value
Champions	22.2	19.2	10,723.5
Loyal Customers	77.5	6.8	2,897.6
Potential Loyalists	178.8	3.0	1,041.8

At-Risk Customers	296.2	1.7	448.5
Can't Lose Them	469.1	1.1	229.8

- Champions are the most valuable group: they buy recently, buy often, and spend the most.
- Loyal Customers are consistent revenue generators and strong candidates for loyalty programs.
- Potential Loyalists show promising behavior and can be converted into Champions with the right nudges.
- At-Risk Customers and Can't Lose Them need win-back and reactivation campaigns before they fully churn.

Cluster Validation Results

The four K-Means clusters closely matched the business segments and helped confirm the behavior patterns observed in RFM scoring.

Cluster	Profile	Recency (days)	Frequency (orders)	Monetary Value
0	Champions	27.5	19.3	10,979.0
1	Loyal / Potential Loyalists	228.3	5.1	2,000.0
2	At-Risk / Can't Lose Them	395.8	1.4	325.0
3	Early-stage Potential Loyalists	28.4	3.0	867.1

Product Preference and Pricing Insights

- Champions buy a diverse mix of higher-value products, which suggests strong cross-sell potential in premium categories.
- Loyal Customers lean toward home décor and furniture, so larger bundles and premium finishes are a natural upsell.
- Potential Loyalists tend to purchase lower-cost novelty items, making them good candidates for frequent low-ticket offers.
- Can't Lose Them and At-Risk Customers often buy seasonal or gift-related items, so reminder-based and seasonal campaigns can help bring them back.
- Champions also show the highest cost per item (around 1,600), while most other groups stay closer to the 100–500 range.

Predictive Modeling Results

A Random Forest Regressor was trained to predict the number of days until the next purchase. The model achieved a Mean Absolute Error of 3.82 days and an R^2 score of 0.39. For highly active customers, the model often predicts very short time gaps, which is useful for timing offers and reminders.

Metric	Value
Mean Absolute Error	3.82 days
R-squared	0.39

Business Recommendations

- Protect Champions with early access, loyalty rewards, and exclusive offers.
- Nurture Loyal Customers with personalized recommendations and bundle-based promotions.
- Convert Potential Loyalists using low-friction offers and regular engagement.
- Run win-back campaigns for At-Risk Customers with timely reminders and targeted discounts.
- Treat Can't Lose Them as a separate reactivation group with seasonal campaigns and stronger incentives.
- Use predicted next-purchase timing to schedule reminders before customers go inactive.

Conclusion

This project shows how RFM analysis, clustering, and predictive modeling can turn raw transaction data into a practical customer strategy. The resulting segments and models provide a clear way to prioritize retention, improve re-engagement, and increase customer lifetime value.

Tech Stack

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook